Topic 055

Geometric Series

Geometric series is one of the most appearing example of series in mathematics and hence it has applications in many areas of science. This is the most basic example. We are choosing this example to understand the concepts of convergence and divergence. So it is very fascinating idea that we are adding infinitely many terms and still if the series is convergent we are going to get some fixed complex number.

Now this example is going to give us this idea in much simple way and we will be able to observe this phenomena that how the addition of infinitely many numbers is going to give us fixed complex number and off course in some cases it is also going to be divergent as well.

Geometric Series: An infinite series of the form $\sum_{n=0}^{\infty} z^n$ is called geometric series.

Examples of geometric series:

z = 1 $\Longrightarrow \sum_{n=0}^{\infty}(1)^n$

z = i $\Longrightarrow \sum_{n=0}^{\infty}(i)^n$

z = 2 + i $\Longrightarrow \sum_{n=0}^{\infty}(2+i)^n$

Now the question is for what values of z geometric series $\sum_{n=0}^{\infty} z^n$ converges or diverges?

Consider the geometric series:

$\sum_{n=0}^{\infty} z^n = 1 , z , z^2 , z^3 , \ldots, z^n, \ldots$

The terms of the series are $1 , z , z^2 , z^3 , \ldots, z^n, \ldots$

Since the ratio of any two consecutive terms is z , so **z is called ratio or common ratio.**

Theorem: If $|z| < 1$, the series $\sum_{n=0}^{\infty} z^n$ converges to $f(z) = \frac{1}{1-z}$.
That is if $|z| < 1$, then

$$\sum_{n=0}^{\infty} z^n = 1 + z + z^2 + z^3 + \ldots + z^n + \ldots = \frac{1}{1-z} .$$

If $|z| \geq 1$, then the series diverges.

Proof: Given $|z| < 1$

We have to show that:

$$\sum_{n=0}^{\infty} z^n = 1 + z + z^2 + z^3 + \ldots + z^n + \ldots = \frac{1}{1-z} .$$

Let $\{S_n\}_{n=1}^{\infty}$ is sequence of partial sums.

$S_n = 1 + z + z^2 + \ldots + z^{n-1}$

$zS_n = z + z^2 + \ldots + z^{n-1} + z^n$

$S_n - zS_n = 1 - z^n$

$S_n = \frac{1-z^n}{1-z}$

$S_n = \frac{1}{1-z} - \frac{z^n}{1-z}$

$$\lim_{n\to\infty} S_n = \lim_{n\to\infty} \frac{1}{1-z} - \lim_{n\to\infty} \frac{z^n}{1-z}$$

$$\lim_{n\to\infty} S_n = \frac{1}{1-z} - 0 = \frac{1}{1-z}$$ *Proved!*

Theorem: *If $\sum_{n=0}^{\infty} z^n$ is a convergent series then $\lim_{n\to\infty} z^n = 0$.*

If $\sum_{n=0}^{\infty} z^n$ is convergent $\Longrightarrow \lim_{n\to\infty} z^n = 0$

$\Longleftrightarrow \quad \lim_{n\to\infty} z^n \neq 0 \quad \Longrightarrow \sum_{n=0}^{\infty} z^n$ is divergent.

We have to show that if n ≥ 1 , then $\sum_{n=0}^{\infty} z^n$ is divergent.

Proof: Given |z| ≥ 1

$\lim_{n\to\infty} |z|^n \neq 0 \quad \Longrightarrow \quad \lim_{n\to\infty} z^n \neq 0$

$\Longrightarrow \sum_{n=0}^{\infty} z^n$ is divergent.

We have proved in much simple way but in fact we have proved non-trivial and important result that when we add infinitely many complex numbers then in some cases it is going to some fixed number and we also calculated the fixed number. In convergent case it is going to be $\frac{1}{1-z}$

Following steps of previous proof we can show that

$$\sum_{n=r}^{\infty} z^n = \frac{z^r}{1-z} \quad \text{if } |z| < 1$$

In general sum of geometric series with common ratio z is

$$\frac{first\ term}{1-z} \quad \text{if } |z| < 1$$

For series:

$\sum_{n=r}^{\infty} az^n$ = a + az + az^2 + ….+ az^n + ….

If |z| < 1 then $\sum_{n=0}^{\infty} az^n = \frac{a}{1-z}$

Corollary: If |z| > 1, the series $\sum_{n=1}^{\infty} z^{-n}$ converges to *f(z)* = $\frac{1}{z-1}$. That is, if |z| > 1 then $\sum_{n=1}^{\infty} z^{-n} = \frac{1}{z} + \frac{1}{z^2} + \ldots + \frac{1}{z^n} + \ldots = \frac{1}{z-1}$

If |z| ≤ 1 then the series diverges.

Example: Evaluate $\sum_{n=4}^{\infty} (\frac{i}{3})^n$

Index Shift: $\sum_{n=4}^{\infty} z_n = z_4 + z_5 + z_6 + \ldots + z_n + \ldots$

The above series can also be written as:

$\sum_{n=0}^{\infty} z_{n+4} = z_4 + z_5 + z_6 + \ldots + z_n + \ldots$

So, we can write it as : $\sum_{n=0}^{\infty} (\frac{i}{3})^{n+4}$

$$(\frac{i}{3})^4 \sum_{n=0}^{\infty} (\frac{i}{3})^n$$

Since, $|\frac{i}{3}| < 1 \Longrightarrow \sum_{n=0}^{\infty} (\frac{i}{3})^n$ converges

Hence,

$\sum_{n=4}^{\infty} (\frac{i}{3})^n$ converges.

Topic 056

d'Alembert's Ratio Test

Theorem: If $\sum_{n=0}^{\infty} \zeta_n$ is a complex series such that

$$\lim_{n\to\infty} \frac{|\zeta_n+1|}{|\zeta_n|} = \mathrm{L}$$

Then the series is:

- Absolutely convergent if L < 1.
- Divergent if L > 1.
- When L = 1 , the test fails and does not say anything about convergence or divergence of the series.

Proof is similar to the case of real series.

Example: Show that the series $\sum_{n=0}^{\infty} \frac{(1-i)^n}{n!}$ converges.

$$\zeta_n = \frac{(1-i)^n}{n!}$$

Now, we have to find:

$$\lim_{n\to\infty} \frac{|\zeta_n+1|}{|\zeta_n|} = \lim_{n\to\infty} \frac{\left|\frac{(1-i)^{n+1}}{(n+1)!}\right|}{\left|\frac{(1-i)^n}{n!}\right|} = \lim_{n\to\infty} \frac{|(1-i)^n(1-i)||n!|}{|n!(n+1)||(1-i)^n|}$$

$$= \lim_{n\to\infty} \frac{|(1-i)|}{n+1} = \lim_{n\to\infty} = \frac{\sqrt{2}}{n+1} = 0$$

L = 0 < 1

Hence, the series $\sum_{n=0}^{\infty} \frac{(1-i)^n}{n!}$ converges.

Example: Show that the series $\sum_{n=0}^{\infty}\frac{(z-i)^n}{2^n}$ converges for all values of z in the disk $|z-i|<2$ and diverges if $|z-i|>2$.

Solution:

$$\zeta_n = \frac{(z-i)^n}{2^n}$$

Now, we have to find:

$$\lim_{n\to\infty}\frac{|\zeta_n+1|}{|\zeta_n|} = \lim_{n\to\infty}\frac{\left|\frac{(z-i)^{n+1}}{2^{n+1}}\right|}{\left|\frac{(z-i)^n}{2^n}\right|} = \lim_{n\to\infty}\frac{|(z-i)^n(z-i)||2^n|}{|2^n.2||(z-i)^n|}$$

$$= \lim_{n\to\infty}\frac{|(z-i)|}{2} = \frac{|(z-i)|}{2} = \mathrm{L}$$

Series will converge if:

$$\mathrm{L} < 1 \Longrightarrow \frac{|(z-i)|}{2} < 1 \Longrightarrow |\, z-i \,| < 2$$

Series will diverge if:

$$\mathrm{L} > 1 \Longrightarrow \frac{|(z-i)|}{2} > 1 \Longrightarrow |\, z-i \,| > 2$$

This kind of example give us support to our idea that we are going to define new complex functions using the concept of series.

Topic 057

Root Test (Part I)

We are going to discuss very powerful test for convergence and divergence of a given series i.e Root Test.

To understand Root Test, we need to understand the concept of Limit Supremum for sequences of real numbers.

In this discussion we will just focus on Limit supremum or simply Lim Sup.

Consider a sequence with terms:

0.5, 1.5, 0.05, 1.05, 0.005, 1.005, 0.0005, 1.0005,…

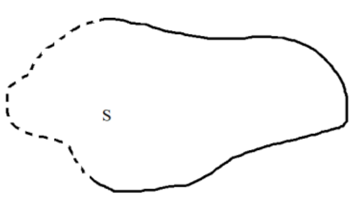
The sequence is divergent, we can not find a limit but can find a limit supremum. Intuitively, we can say that 1 is the limit supremum of this sequence.

In fact, 0 and 1 are limit points of this sequence.

Recall:

Accumulation Point / Limit Point:

A point z_0 is called an accumulation point of a set S if every deleted neighborhood of z_0 contains at least one point of S.



In given sequence, {0,1} is the set of limit points. Supremum in this limit set is 1 so 1 is the limit supremum of this sequence.

Definition (limit supremum)

Let $\{t_n\}$ be a sequence of positive real numbes. The limit supremum of the sequence, denoted by:

$$\lim_{n\to\infty} sup\ t_n$$

is the smallest real number L having the property that for any

$\varepsilon > 0$, there are at most finitely many terms in the sequence that are larger than L + ε. If there is no such L then

$$\lim_{n\to\infty} sup\ t_n = \infty$$

Example: Consider a sequence

$\{t_n\} = \{1,2,3,1,2,3,1,2,3,1,2,3,1,2,3,\ldots.\}$ is diverging because it is oscillating between 1, 2 and 3. But we can calculate lim sup.

$\lim_{n\to\infty} sup\ t_n = 3$

Example: Consider a sequence

$\{t_n\} = \{1,2,3,4,5,6,7,8,\ldots.\}$ is diverging.

In this case, $\lim_{n\to\infty} sup\ t_n = \infty$

Relation between limit supremum and limit of a sequence:

If a sequence $\{t_n\}$ is convergent and converging to M that is:

$$\lim_{n\to\infty} t_n = M$$

Then we say that:

$$\lim_{n\to\infty} sup\ t_n = M$$

Topic 058

Root Test (Part II)

Theorem (Root Test): *Suppose the series* $\sum_{n=0}^{\infty} \zeta_n$ *has*

$$\lim_{n\to\infty} sup|\zeta_n|^{\frac{1}{n}} = L$$

Then the series:

- *Absolutely convergent if L < 1.*
- *Divergent if L > 1.*

It does not say anything when L = 1.

Proof: *Assume that* $\lim_{n\to\infty} sup|\zeta_n|^{\frac{1}{n}} = L$ *and L < 1.*

We need to show that the series $\sum_{n=0}^{\infty} \zeta_n$ *converges absolutely.*

We can find r such that L < r < 1.

By definition of limit supremum there are only finitely many terms of the sequence $\{|\zeta_n|^{\frac{1}{n}}\}$ *greater than r. That is there is some integer N such that*

$$|\zeta_n|^{\frac{1}{n}} < r \qquad \text{for } n \geq N$$

So,

$$|\zeta_n| < r^{n} \qquad \text{for } n \geq N$$

Recall:

Theorem: If $|z| < 1$, the series $\sum_{n=0}^{\infty} z^n$ converges to $f(z) = \frac{1}{1-z}$. That is if $|z| < 1$, then

$\sum_{n=0}^{\infty} z^n = 1 + z + z^2 + z^3 + \ldots + z^n + \ldots = \frac{1}{1-z}$.

If $|z| \geq 1$, then the series diverges.

In this case, the geometric series $\sum_{n=0}^{\infty} r^n$ *converges since* $r < 1$.

Since, $|\zeta_n| < r^n$ *for* $n \geq N$

So, by comparison test the series $\sum_{n=N+1}^{\infty} |\zeta_n|$ *converges and does the following* $\sum_{n=0}^{\infty} |\zeta_n|$.

Since absolute convergence implies convergence so the following series converges.

$\sum_{n=N+1}^{\infty} \zeta_n \implies \sum_{n=0}^{\infty} \zeta_n$ *converges.*

Now,

Assume that $\lim_{n\to\infty} sup|\zeta_n|^{\frac{1}{n}} = L$ *and* $L > 1$.

We need to show that the series $\sum_{n=0}^{\infty} \zeta_n$ *diverges.*

Since $1 < L$ *so there exist* r *such that* $1 < r < L$

By definition of limit supremum there exist N *such that*

$$|\zeta_n|^{\frac{1}{n}} > r \; for \; n \geq N$$

which implies $|\zeta_n| > r^n \; for \; n \geq N$

As $r > 1$ *so,*

$$\lim_{n\to\infty} |\zeta_n| \neq 0$$

Hence, $\lim_{n\to\infty} \zeta_n \neq 0$

which implies $\sum_{n=0}^{\infty} \zeta_n$ *diverges.* ***Proved!***

Example: *Check whether the following series converges or diverges.*

$$\sum_{n=0}^{\infty} \left(\frac{1+i}{2}\right)^n$$

Solution:

$\zeta_n = \left(\frac{1+i}{2}\right)^n$

$$\lim_{n\to\infty} sup|\zeta_n|^{\frac{1}{n}} = \lim_{n\to\infty} sup\left|\left(\frac{1+i}{2}\right)^n\right|^{\frac{1}{n}}$$

$$= \lim_{n\to\infty} sup \left|\frac{1+i}{2}\right|$$

$$= \left|\frac{1}{2} + \frac{1}{2}i\right| = \sqrt{\frac{1}{4} + \frac{1}{4}}$$

$$= \frac{1}{\sqrt{2}} = 0.707 = L < 1$$

$\Longrightarrow$ *The given series converges.*

Topic 059

Power Series Functions

(Part I)

An infinite series is an expression of the form $\sum_{n=1}^{\infty} a_n$

Consider an infinite series of the form $\sum_{n=1}^{\infty} c_n\,(z-\alpha)^n$ *where* c_n *and* α *are fixed complex numbers.*

Warning:

The series $\sum_{n=1}^{\infty} c_n\,(z-\alpha)^n$ *defines a function only if it converges.*

Power Series Funcion *is given by:*

$$\sum_{n=1}^{\infty} c_n\,(z-\alpha)^n$$

Domain*= collection of all points z for which the series converges.*

Example:

$$\alpha = 3\,, c_n = n!$$

$$f(z) = \sum_{n=1}^{\infty} n!\,(z-3)^n$$

Example:

$$\alpha = 0\,, c_n = 1$$

$$f(z) = \sum_{n=1}^{\infty} (z)^n$$

Notation:

$\sum_{n=1}^{\infty} c_n (z-\alpha)^n = c_0(z-\alpha)^0 + c_1(z-\alpha)^1 + \ldots + c_n(z-\alpha)^n + \ldots$

But 0^0 *is undefined.*

So, we agree on the following convention:

$\sum_{n=1}^{\infty} c_n (z-\alpha)^n = c_0 + c_1(z-\alpha)^1 + \ldots + c_n(z-\alpha)^n + \ldots$

In this discussion, we see how we can use an infinite series to define complex valued function.

Topic 060

Power Series Functions

(Part II)

Nothing is explained in this module.

Topic 061

Power Series Functions

(Part III)

In this module, we are going to continue our discussion in finding the domain. In fact we are going to calculate different radius of convergence.

Theorem: *Suppose that* $\sum_{n=1}^{\infty} c_n (z-\alpha)^n$. *Then the set of points z for which the series converges is one of the following:*

1. *The single point z =* α.
2. *The disk* $D_p(\alpha) = \{z: |z - \alpha| < \rho\}$, *along with part (either none, some or all) of the circle* $C_p = \{z: |z - \alpha| = \rho\}$
3. *The entire complex plane.*

Radius of convergence: *The number* ρ *such that the series* $\sum_{n=1}^{\infty} c_n (z-\alpha)^n$ *converges for all z satisfying* $|z - \alpha| < \rho$ *and diverges for all z satisfying* $|z - \alpha| > \rho$ *is called radius of convergence.*

Question: *How to find radius of convergence?*

Theorem: *For the power series function* $f(z) = \sum_{n=1}^{\infty} c_n (z-\alpha)^n$ *we can find its radius of convergence denoted by* ρ *using any of the following methods:*

1. *Cauchy's root Test*

$$\rho = \frac{1}{\lim_{n\to\infty} |c_n|^{\frac{1}{n}}}$$

$(Provided\ the\ limit\ exist)$

2. *Cauchy-Hadamard root Test*

$$\rho = \frac{1}{\lim_{n\to\infty} sup\, |c_n|^{\frac{1}{n}}}$$

$(\ the\ limit\ always\ exist)$

3. *d' Alembert's ratio test*

$$\rho = \frac{1}{\lim\limits_{n\to\infty}\left|\frac{c_{n+1}}{c_n}\right|}$$

$(Provided\ the\ limit\ exist)$

Example: *Find radius of convergence of following series:*

$$\sum_{n=0}^{\infty}\left(\frac{n+2}{3n+1}\right)^n (z-4)^4$$

Solution: *According to Cauchy's root test, the radius of convergence is:*

$$\rho = \frac{1}{\lim\limits_{n\to\infty}|c_n|^{\frac{1}{n}}}$$

So we calculate the following limit:

$$\lim_{n\to\infty}|c_n|^{\frac{1}{n}}$$

Where $c_n = \left(\frac{n+2}{3n+1}\right)^n$

$$= \lim_{n\to\infty}\left|\left(\frac{n+2}{3n+1}\right)^n\right|^{\frac{1}{n}}$$

$$= \lim_{n\to\infty}\frac{1+\frac{2}{n}}{3+\frac{1}{n}} = \frac{1}{3}$$

Hence radius of convergence is:

$$\rho = \frac{1}{\frac{1}{3}} = 3$$

Example: *Find radius of convergence of following series:*

$\sum_{n=1}^{\infty} c_n z^n = 4z + 5^2z^2 + 4^3z^3 + 5^4z^4 + \ldots..$

Solution:

In this case

$\{|c_n|^{\frac{1}{n}}\} = \{4,5,4,5,4,5,\ldots\}$

So, the limit

$$\lim_{n\to\infty} |c_n|^{\frac{1}{n}}$$

does not exist.

Hence we use Cauchy-Hadamard root test.

So the limit:

$$\lim_{n\to\infty} sup|c_n|^{\frac{1}{n}} = 5$$

Hence radius of convergence is $\rho = \frac{1}{5}$

Example: *Find radius of convergence of following series:*

$$\sum_{n=1}^{\infty} \frac{1}{n!} z^n$$

Solution: *According to d' Alembert's ratio test, the radius of convergence is:*

$$\rho = \frac{1}{\lim\limits_{n\to\infty}\left|\frac{c_{n+1}}{c_n}\right|}$$

where, in this example:

$$c_n = \frac{1}{n!}$$

We calculate the following limit:

$$\lim_{n\to\infty}\left|\frac{c_{n+1}}{c_n}\right|$$

$$= \lim_{n\to\infty}\left|\frac{\frac{1}{(n+1)!}}{\frac{1}{n!}}\right|$$

$$= \lim_{n\to\infty}\left|\frac{1}{n+1}\right| = 0$$

$$\rho = \frac{1}{\lim\limits_{n\to\infty}\left|\frac{c_{n+1}}{c_n}\right|}$$

$$\rho = \frac{1}{0} = \infty$$

So, the radius of convergence is $\rho = \infty$

In this discussion, we have seen some ways for calculating the radius of convergence of given power series function. And we have considered some examples. These ways are going to help us in future when we are going to construct new examples of complex valued functions. And we will see that using this method we can easily calculate the domain of a given complex valued function.

Topic 062

Power Series Functions

Term by term differentiation

(Part I)

Now, we are interested in some other properties like differentiation and integration of power series functions.

Question: *When can we differentiate the power series function?*

What is the derivative, if exist?

Notation: *For a function f(z) we denote by $f^{(k)}$ the kth derivative of f(z). When k = 0, $f^{(k)}$ denotes the function itself, that is $f^{(0)}(z) = f(z)$.*

Theorem: *Suppose that the function f(z) = $\sum_{n=0}^{\infty} c_n (z-\alpha)^n$ has radius of convergence $\rho > 0$. Then* *(i)* *f(z) is infinitely differentiable for all* $z \in D_p(\alpha)$.

(ii) *for all k*

$$f^{(k)}(z) = \sum_{n=k}^{\infty} n(n-1)(n-2)\ldots(n-k+1)c_n(z-\alpha)^{n-k}$$

(iii) $c_k = \frac{f^{(k)}(\alpha)}{k!}$

Proof: *We first prove the second part that is:*

(ii) *for all k*

$$f^{(k)}(z) = \sum_{n=k}^{\infty} n(n-1)(n-2)\ldots(n-k+1)c_n(z-\alpha)^{n-k}$$

In the first step, we prove the above statement for k = 1. In the second step we prove it for any k.

First step k = 1: *We define*

$g(z) = \sum_{n=1}^{\infty} nc_n (z-\alpha)^{n-1}$

$S_j(z) = \sum_{n=0}^{j} c_n (z-\alpha)^n$, $R_j(z) = \sum_{n=j+1}^{\infty} c_n (z-\alpha)^n$

Here

- *$S_j(z)$ is simply is the (j + 1)st partial sum of the series f(z),*
- *$R_j(z)$ is the sum of the remaining terms of that series.*

The radius of convergence of $g(z) = \sum_{n=1}^{\infty} nc_n (z-\alpha)^{n-1}$ *is* ρ.

According to d'Alembert's ratio test:

$$\rho = \frac{1}{\lim_{n\to\infty} \left|\frac{c_{n+1}}{c_n}\right|}$$

For g(z):

$$\lim_{n\to\infty}\left|\frac{c_{n+1}}{c_n}\right| = \lim_{n\to\infty}\left|\frac{(n+1)c_{n+1}}{nc_n}\right| = \lim_{n\to\infty}\left|\frac{n+1}{n}\right| \lim_{n\to\infty}\left|\frac{c_{n+1}}{c_n}\right|$$

$$= \lim_{n\to\infty}\left(1+\frac{1}{n}\right)\lim_{n\to\infty}\left|\frac{c_{n+1}}{c_n}\right| = \frac{1}{\rho}$$

For fixed $z_0 \in D_p(\alpha)$ we show that $f'(z_0) = g(z_0)$

That is, we want to show that:

$$f'(z_0) = \lim_{z\to z_0}\frac{f(z) - f(z_0)}{z - z_0} = g(z_0)$$

Recall the definition of limit:

$\lim_{z\to z_0} \mathrm{f(z)} = L$ *is*

" for every $\varepsilon > 0$ there exist a $\delta > 0$ such that $|f(z) - L| < \varepsilon$

whenever $0 < |z - z_0| < \delta$"

Hence we need to show the following:

for every $\varepsilon > 0$ there exist a $\delta > 0$ such that

$\left|\frac{f(z) - f(z_0)}{z - z_0} - g(z0)\right| < \varepsilon$ *whenever $0 < |z - z_0| < \delta$ and*

$z \in D_p(\alpha)$.

Let $\varepsilon > 0$ and $z_0 \in D_p(\alpha)$. Choose $r < \rho$ such that $z_0 \in D_r(\alpha)$. We want to find δ small enough such that

$D_\delta(z_0) \subset D_r(\alpha) \subset D_p(\alpha)$

and other conditions that we impose during the proof.

Since f(z) = S_j(z) + R_j(z), so

$$\left[\frac{f(z)-f(z_0)}{z-z_0}-g(z0)\right]=\left[\frac{S_j(z)+R_j(z)-(S_j(z_0)+R_j(z_0))}{z-z_0}-g(z0)\right]$$

$$=\left[\frac{S_j(z)-S_j(z_0)}{z-z_0}+\frac{R_j(z)-R_j(z_0)}{z-z_0}-g(z0)\right]$$

$$=\left[\frac{S_j(z)-S_j(z_0)}{z-z_0}-S_j^{\prime(z_0)}\right]+\left[S_j{}'(z_0)-g(z0)\right]+\left[\frac{R_j(z)-R_j(z_0)}{z-z_0}\right]$$

$$\left|\frac{f(z)-f(z_0)}{z-z_0}-g(z0)\right|=\left|\left[\frac{S_j(z)-S_j(z_0)}{z-z_0}-S_j^{\prime(z_0)}\right]+\left[S_j{}'(z_0)-g(z0)\right]+\left[\frac{R_j(z)-R_j(z_0)}{z-z_0}\right]\right|$$

$$\left|\frac{f(z)-f(z_0)}{z-z_0}-g(z0)\right|\le\left|\frac{S_j(z)-S_j(z_0)}{z-z_0}-S_j^{\prime(z_0)}\right|+\left|S_j{}'(z_0)-g(z0)\right|+\left|\frac{R_j(z)-R_j(z_0)}{z-z_0}\right|$$

$$\left|\frac{R_j(z)-R_j(z_0)}{z-z_0}\right|=\left|\frac{1}{z-z_0}\sum_{n=j+1}^{\infty}c_n(z-\alpha)^n-\sum_{n=j+1}^{\infty}c_n(z_0-\alpha)^n\right|$$

$$=\left|\frac{1}{z-z_0}\sum_{n=j+1}^{\infty}\left(c_n(z-\alpha)^n-c_n(z_0-\alpha)^n\right)\right|$$

$$\le\sum\nolimits_{n=j+1}^{\infty}|c_n|\left|\frac{(z-\alpha)^n-(z_0-\alpha)^n}{z-z_0}\right|$$

Now,

$$\left|\frac{(z-\alpha)^n-(z_0-\alpha)^n}{z-z_0}\right|<nr^{n-1}$$

Hint:

$$\left|\frac{p^n - q^n}{p - q}\right| = |p^{n-1} + p^{n-2}q + p^{n-3}q^2 + \cdots + pq^{n-2} + q^{n-1}|$$

$$\leq |p^{n-1}| + |p^{n-2}q| + |p^{n-3}q^2| + .. + |pq^{n-2}| + |q^{n-1}|$$

Let $p = z-\alpha$, $q = z_0 - \alpha \Longrightarrow p - q = z - z_0$

Also, $|\, z-\alpha \,| < r$, $|z_0 - \alpha| < r$

Hence,

$\left|\frac{R_j(z)-R_j(z_0)}{z-z_0}\right| \leq \sum_{n=j+1}^{\infty}|c_n|\; nr^{n-1}$

Now,

$\sum_{n=j+1}^{\infty}|c_n|\; nr^{n-1}$ *converges?*

Let's apply d'Alembert's ratio test:

$$\rho = \frac{1}{\lim_{n\to\infty}\left|\frac{c_{n+1}}{c_n}\right|}$$

$$\lim_{n\to\infty}\left|\frac{c_{n+1}}{c_n}\right| = \lim_{n\to\infty}\left|\frac{(c_{n+1})(n+1)r^{n-1+1}}{c_n n r^{n-1}}\right|$$

$$= r \lim_{n\to\infty}\left|\frac{(c_{n+1})}{c_n}\right| \lim_{n\to\infty}\frac{n+1}{n}$$

$$= r\,\frac{1}{\rho} < 1 \qquad\qquad \therefore r < \rho$$

Hence, $\lim_{n\to\infty}\left|\frac{c_{n+1}}{c_n}\right|$ *converges.*

Since, $\sum_{n=j+1}^{\infty}|c_n|\, nr^{n-1}$ *converges, so we can find* N_1 *such that:*

$\sum_{n=j+1}^{\infty}|c_n|\, nr^{n-1} < \frac{\varepsilon}{3} \qquad \forall\, j \geq N_1$

Topic 063

Continuation of proof

Term by term differentiation

Now, we will focus on $\left|S_j{}'(z_0) - g(z_0)\right|$.

Given:

$g(z) = \sum_{n=1}^{\infty} nc_n(z-\alpha)^{n-1}$ *and* $S_j(z) = \sum_{n=0}^{j} c_n(z-\alpha)^n$

Since, $S_j(z)$ *is a polynomial. So,*

$S_j{}'(z) = \sum_{n=1}^{j} nc_n(z-\alpha)^{n-1}$

Hence, $\lim_{j\to\infty} S_j{}'(z) = g(z)$

which implies:

$\lim_{j\to\infty} S_j{}'(z_0) = g(z_0)$

Using definition of limit we get:

$\exists\, N_2$ such that $\left|S_j(z_0) - g(z_0)\right| < \frac{\varepsilon}{3}$ for all j

Now:

$$\left[\frac{S_j(z) - S_j(z_0)}{z - z_0} - S_j'^{(z_0)}\right]$$

Since $S_j(z)$ is a polynomial so it is differentiable.

(Using definition of derivative)

Hence for $\frac{\varepsilon}{3} > 0$ there exist $\delta > 0$ such that:

$\left[\frac{S_j(z)-S_j(z_0)}{z-z_0} - S_j'^{(z_0)}\right] < \frac{\varepsilon}{3}$

Whenever $z \in D_p(\alpha)$ *and* $0 < |z - z_0| < \delta \;\; \forall\, j \geq \max\{N_1, N_2\}$

Hence,

$\left|\frac{f(z)-f(z_0)}{z-z_0} - g(z0)\right| \leq \left|\frac{S_j(z)-S_j(z_0)}{z-z_0} - S_j'^{(z_0)}\right| + \left|S_j{}'(z_0) - g(z0)\right| + \left|\frac{R_j(z)-R_j(z_0)}{z-z_0}\right|$

$\left|\frac{f(z)-f(z_0)}{z-z_0} - g(z0)\right| \leq \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon$

$\forall\, j \geq \max\{N_1, N_2\}$ *and* $z \in D_\delta(z_0) \subseteq D_p(\alpha)$

So we have done for the first derivative!

In the ***Second step*** *we prove it for any k.*

$f^{(1)}(z) = \sum_{n=1}^{\infty} nc_n\,(z-\alpha)^{n-1}$

$g(z) = \sum_{n=2}^{\infty} n(n-1)c_n\,(z-\alpha)^{n-2}$

proceeding in the same way, we can prove for any k.

For proving the third part of the theorem i.e $c_k = \frac{f^{(k)}(\alpha)}{k!}$ *, put* $z = \alpha$ *in* $f^{(k)}(z)$.

Topic 064

Power Series Functions

Term by term differentiation

Part (II)

This is a continuation of our discussion on differentiation of power series functions. Now remember we are using power series to define functions and the question is how can we calculate the derivative of these functions and what is the derivative?

<u>Power Series Funcion</u> *is given by:*

$$\sum_{n=1}^{\infty} c_n \, (z-\alpha)^n$$

Domain*= collection of all points z for which the series converges.*

We know that the function is well defined when the series converges.